Advanced Microeconomics I
Note 1: Preference and choice

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Introduction

- Individual decision making
- Suppose that X is a set of *alternatives*, and an agent must choose from this set (or a subset of X).
- Two approaches to model an agent's decision making:
  - Preference-based approach
  - Choice-based approach

Preferences - binary relations

- Let $X \times X$ denote the *Cartesian product* of all ordered pairs:

$$X \times X = \{(x, y) : x \in X, y \in X\}$$

- A **binary relation** B on X is a subset of $X \times X$, i.e., $B \subseteq X \times X$.
- If $(x, y) \in B$, then write xBy.
- If $(x, y) \notin B$, then write $x\bar{B}y$.

Some properties of a binary relation

A binary relation B on X is

- **reflexive** if xBx for all $x \in X$.
- **irreflexive** if $x\bar{B}x$ for all $x \in X$.
- **symmetric** if xBy implies yBx.
- **asymmetric** if xBy implies $y\bar{B}x$.
- **transitive** if xBy and yBz imply xBz.
- **negatively transitive** if $x\bar{B}y$ and $y\bar{B}z$ imply $x\bar{B}z$.
- **complete** if for all $x, y \in X$, xBy or yBx (*or both*).

Preferences

- There are various ways of defining /modeling preference relations.
- First, consider the "P-model".
- The *primitive* of the model is a binary relation P on X, and P is interpreted as the "strictly better than" relation.
- We want to make sure that the preferences are "rational" or "consistent".
- We impose two conditions on the strict preference relation P:
  - P is asymmetric: if x is strictly better than y, then y is not strictly better than x.
  - P is negatively transitive: if x is not strictly better than y and y is not strictly better than z, then x is not strictly better than z.
- Did we require too little?

Proposition If P is asymmetric and negatively transitive, then

(1) P is irreflexive

(2) P is transitive

(3) xPy and $z\bar{P}y$ imply xPz; $y\bar{P}x$ and yPz imply xPz

- Next, consider the "$\succeq$-model".
- In this case, the primitive of the model is a binary relation $\succeq$ on X, and $\succeq$ is interpreted as the "weakly better than" relation.
- We require $\succeq$ to be *complete* and *transitive*.
- It can be shown that if $\succeq$ is complete and transitive, then it is reflexive and negatively transitive.

The "P-model" and the "$\succeq$-model" are equivalent.

Proposition

(i) Given the asymmetric and negatively transitive P, define a binary relation $\succeq'$ on X as follows: for any $x, y \in X$, $x \succeq' y$ if $y\bar{P}x$. Then $\succeq'$ is complete and transitive.

(ii) Given the complete and transitive $\succeq$, define a binary relation P' on X as follows: for any $x, y \in X$, $xP'y$ if $x \succeq y$ and $y \not\succeq x$. Then P' is asymmetric and negatively transitive.

Proof of (i). Completeness: Consider any $x, y \in X$. If xPy, then by the asymmetry of P, we have $y\bar{P}x$. Hence by the definition of $\succeq'$, $x \succeq' y$. If $x\bar{P}y$, then by the definition of $\succeq'$, $y \succeq' x$.

Transitivity: Consider any $x, y, z \in X$ with $x \succeq' y$ and $y \succeq' z$. By the definition of $\succeq'$, $y\bar{P}x$ and $z\bar{P}y$. Then by the negative transitivity of P, $z\bar{P}x$. It follows that $x \succeq' z$.

Proof of (ii). Asymmetry is obvious.

Negative transitivity: Consider any $x, y, z \in X$ with $x\bar{P}'y$ and $y\bar{P}'z$. Suppose that $y \not\succeq x$. Then by completeness of $\succeq$, $x \succeq y$. Hence by the construction of P', $xP'y$, contradiction. So we have $y \succeq x$. By a similar argument, it can be shown that $z \succeq y$. By the transitivity of $\succeq$, $z \succeq x$. Given the construction of P', it follows that $x\bar{P}'z$. □

- From now on, we use the $\succeq$-model.
- Define a **preference relation** on X as a binary relation $\succeq$ on X. The preference relation $\succeq$ is **rational** if it is complete and transitive.
- Given a preference relation $\succeq$ on X,
  - denote its "asymmetric component" as $\succ$, i.e., $x \succ y$ if $x \succeq y$ but $y \not\succeq x$.
  - denote its "symmetric component" as $\sim$, i.e., $x \sim y$ if $x \succeq y$ and $y \succeq x$.

- More on rationality
- Completeness: can you always compare?
  - Suppose that I offer you a trip to the moon, do you want to go to the northern part or the southern part?
- Two common sources of intransitivity:
  - Aggregation
  - The use of similarities

Choice correspondence - the weak axiom of revealed preference

- A full description of an agent's choice behavior in all possible scenarios.
- Let $\mathcal{D}$ be a collection of non-empty subsets of X.
  - Notice that $\mathcal{D}$ may not include *all* the subsets of X.
- C is a **choice correspondence** if for any $A \in \mathcal{D}$, $C(A) \subseteq A$ and $C(A) \neq \phi$.
- A choice correspondence C satisfies the **weak axiom of revealed preference** (WARP) if the following is true: if for some $A \in \mathcal{D}$ with $x, y \in A$ we have $x \in C(A)$ and $y \notin C(A)$, then for any $B \in \mathcal{D}$ with $x, y \in B$ we must have $y \notin C(B)$.
  - If, in some case, x is chosen over y, then y should never be chosen in the presence of x.
- An equivalent definition: C satisfies WARP if the following is true: if for some $A \in \mathcal{D}$ with $x, y \in A$ we have $x \in C(A)$, then for any $B \in \mathcal{D}$ with $x, y \in B$ and $y \in C(B)$ we must have $x \in C(B)$.
  - If, in some case, x is chosen in the presence of y, then y should never be chosen over x.
- The *richness* of the domain $\mathcal{D}$ is important.

- *Sometimes*, WARP can be decomposed into the following two conditions on a choice correspondence C.
- **Sen's property** α: given $A, B \in \mathcal{D}$, if $x \in A \subseteq B$ and $x \in C(B)$, then $x \in C(A)$.
  - Amartya Sen's paraphrase of this: if the world champion in some game is a Pakistani, then he must also be the champion of Pakistan.
- **Sen's property** β: given $A, B \in \mathcal{D}$, if $A \subseteq B$, $x \in C(A), y \in C(A)$ and $x \in C(B)$, then $y \in C(B)$.
  - Sen's paraphrase: if the world champion in some game is a Pakistani, then all champions (in this game) of Pakistan are also world champions.
- WARP implies Sen's properties α and β.
- If $\mathcal{D}$ includes at least all the subsets of X of size 2, then Sen's properties α and β imply WARP.
- If for any $A, B \in \mathcal{D}$ we have $A \cap B \in \mathcal{D}$, then Sen's properties α and β imply WARP.

From preferences to choice correspondence

- Given a preference relation $\succeq$ on X, an *induced* correspondence is $C_{\succeq}$: for any $A \in \mathcal{D}$, $C(A) = \{x \in A : x \succeq y, \forall y \in A\}$.
- Assume that X is finite. If $\succeq$ is rational, then $C_{\succeq}$ is a well-defined choice correspondence that satisfies WARP.

Rationalizing

A choice correspondence C can be **rationalized** if there exists a rational preference relation $\succeq$ on X such that $C = C_{\succeq}$, i.e., $C(A) = C_{\succeq}(A)$ for all $A \in \mathcal{D}$.

Proposition. *Suppose that $\mathcal{D}$ includes at least all subsets of X of size up to 3, and $|C(A)| = 1$ for all $A \in \mathcal{D}$ (i.e., C is a "choice function"). Then C can be rationalized if and only if C satisfies Sen's property α.*

Proof. "Only if" part: if C can be rationalized, then there exists rational $\succeq$ such that $C = C_{\succeq}$. From the previous discussion, we know that $C_{\succeq}$ satisfies WARP, hence Sen's property α.

"If" part. Define $\succeq$ on X as follows: for any $x, y \in X$, let $x \succeq y$ if $\{x\} = C(\{x, y\})$.

First, we show that $\succeq$ is rational. Consider any $x, y \in X$. We have $x \succeq y$ if $\{x\} = C(\{x, y\})$, $y \succeq x$ if $\{y\} = C(\{x, y\})$. So $\succeq$ is complete. Now, suppose that $\succeq$ is not transitive. Then there exist $x, y, z \in X$ such that $x \succeq y$, $y \succeq z$ and $x \not\succeq z$. It follows that $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$ and $C(\{x, z\}) = \{z\}$. Then consider the set $\{x, y, z\}$. Given that C satisfies Sen's property α, we have: $C(\{x, y\}) = \{x\}$ implies $C(\{x, y, z\}) \neq \{y\}$, $C(\{y, z\}) = \{y\}$ implies $C(\{x, y, z\}) \neq \{z\}$, and $C(\{x, z\}) = \{z\}$ implies $C(\{x, y, z\}) \neq \{x\}$. That is, $C(\{x, y, z\}) = \phi$, contradiction.

It remains to show that $C = C_{\succeq}$. Suppose that for some $A \in \mathcal{D}$, $C(A) = \{x\} \neq \{y\} = C_{\succeq}(A)$. By the construction of $C_{\succeq}$, $y \succeq x$. Then by the construction of $\succeq$, $\{y\} = C(\{x, y\})$. However, $\{x, y\} \subseteq A$ and $C(A) = \{x\}$, contradicting to Sen's property α. □

Proposition. *Suppose that $\mathcal{D}$ includes at least all subsets of X of size up to 3, and C is a choice correspondence. C can be rationalized if and only if C satisfies the weak axiom of revealed preference.*